INCH-POUND

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06 July 1993
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4 November 1975

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY

TARGET ACCELERATION BOARD: 11747207

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for target acceleration printed circuit board assembly, hereinafter identified as target acceleration board (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-T-28800 - Test Equipment for use with Electrical and Electronic Equipment, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-I-45607 Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings, and publications. The following other Government documents, drawings, and publications-form a part of this document to the-extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT' AND ENGINEERING CENTER (ARDEC)

10554938 - Connector
11747207 - Target Acceleration Board
11747209 - Schematic, Target Acceleration Board
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

PACKAGING DATA SHEET

PS11747207 - Packaging of Target Acceleration Board

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11747207, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the target acceleration board shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The target acceleration board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to-sinusoidal vibrations, in each of three mutually perpendicular axes, as specified in MIL-T-28800 for Type 1, Class 3, Style F test equipment.

3.3.2 Shock. The target acceleration board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The target acceleration board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The target acceleration board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32 deg. F and +131 deg. E'.

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3.3.4.2 Non-operating (storage). The target acceleration board shall not suffer permanent damage when subjected to the constant temperature extremes of -60 Degree F for a period of 24 hours and +160 Degree F for a period of 4 hours (after temperature stabilization) , and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The target acceleration board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be continuous electrically as specified on the schematic diagram 11747209.

3.4.2 Resistance. Resistance shall be 1K ohms ±10% between the following terminals.

From	To
P1-37	P1-30
P1-39	P1-30
P1-7	P1-30
Pi-l	P1-30

3.5 Performance. For the requirements of 3.5.1 through 3.5.3 logic levels shall be as follows:

Logic 0 = 0.0 ± 0.5 Vdc

Logic 1 = 4.25 ± 1.00 vdc

↓ = momentary transition from logic 1 to logic 0.

3.5.1 Target signals. When supplied with the inputs as specified in Table I, the target acceleration board shall yield the outputs as specified in Table I.

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TABLE I. Target signals, input/output characteristics,

REQUIREMENT	CHARACTERISTICS	TERMINALS
Input Power	5.00 ± 0.25 Vdc regulated, AC ripple not greater than 50 mv p-p, 600 ma capacity	P1-30(+) P1-24 (Ret.)
Input Signals		
T_o^1	Positive pulse from 0 to 4.0 ±0.5 vdc, width of between 0.2 to 0.6 us at 50% points, PRF of 20.000 ±0.200 KHz	P1-5 P1-24 (Ret.)
Target Overflow	Positive pulse from 0 to 4.0 ±0.5 Vdc, width of 1.0 ±0.1 US, leading edge delayed 15.0 ±1.0 us from leading edge of T_o^1	P1-17 P1-24 (Ret.)
Output Signals		
Target Clear	Positive pulse from 0 to 4.0 ±0.5, Vdc, width of 15.0 ±1.0 us trailing edge is coincident with leading edge of T_0^1, PRF of 20,000 ±0.200 KHz	P1-15 P1-24 (Ret.)
Target Trigger	Positive pulse from 0 to 4.0 ±0.5 Vdc, width of 35.0 ±1.0 us, leading edge coincides with leading edge of T_o^1, PRF of 20.000 ±0.200 KHz	Pi-lo P1-24 (Ret.)
TARGET RESET	Positive pulse from 0 to 4.0 ±0.5, Vdc, width of 5.2±0.5 us, leading edge delayed 15.0 ±1.0 us from leading edge of T_o^1, PRF of 20.000 ±0.200 KHz	P1-19 P1-24 (Ret.)
Strobe	Positive pulse from 0 to 4.0 ±0.5 Vdc, width of 1.4±0.4 us, leading edge delayed 42.0±2.0 us from leading edge of T_o^1, PRF of 20.000 ± 0.200 KHz	P1-13 P1-24 (Ret.)

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TABLE I. Target signals, input/output characteristics. (cont.)

REQUIREMENT	CHARACTERISTICS	TERMINALS
Doppler	50% square wave, from 0 to 4.0 ±0.5, vdc, PRF of 1.000 ±0.010 KHz	P1-35 P1-24 (Ret.)
80 M/S ^{2}	50% square wave, 0 to 4.0 ± 0.5 Vdc, PRF 31.25 ± 1.00 HZ	P1-44 P1-24 (Ret.)
20 M/S ^{2}	50% square wave, 0 to 4.0 ± 0.5 Vdc, PRF of 7.80 ± 0.05 HZ	P1-42 P1-24 (Ret.)

3.5.2 Count-up, count-down outputs. With the input Logic Levels applied sequentially (Table II - left to right per test, tests 1 to 8) as specified in Table II, and a ↓ applied to Pi-l to start each test, the output signals shall be as specified in Table II. Any output signal shall be changed to Logic 0 by applying a ↓ to P1-52. Between the time a ↓ is applied to Pi-l and P1-52, P1-55 shall be at Logic 0 and at any other time Logic 1. The 20 M/S ^{2} and 80 M/S ^{2} signals applied to P1-40 are obtained from P1-42 and P1-44 respectively.

3.5.3 stop, $\overline{\text{stop}}$ Outputs. With the input Logic Levels applied sequentially (Table III - left to right per test, tests 1 to 7) as specified in Table III, the output signals shall be as specified in Table III.

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TABLE II. Count-up, count-down output.

	INPUT SIGNALS						OUTPUT SIGNALS	
Test	P1-7	P1-9	P1-11	P1-37	P1-39	P1-40	P1-21	P1-23
1	Logic 1	Logic 0	Logic 1	Logic 1	Logic 0	Logic 1	Fig. 1	Logic
2	Logic 1	Logic 1	Logic 0	Logic 1	Logic 0	Logic 1	Logic 1	Fig. 1
3	Logic 1	Logic 0	Logic 1	Logic 0	Logic 1	Logic 1	Fig. 2	Logic
4	Logic 1	Logic 1	Logic 0	Logic 0	Logic 1	Logic 1	Logic 1	Fig. 2
5	Logic 0	Logic 0	Logic 1	Logic 1	Logic 1	20 M/S^2	Fig. 3	Fig. 3
6	Logic 0	Logic 1	Logic 0	Logic 1	Logic 1	20 M/S^2	Fig. 3	Fig. 3
7	Logic 0	Logic 0	Logic 1	Logic 1	Logic 1	80 M/S^2	Fig. 4	Fig. 4
8	Logic 0	Logic 1	Logic 0	Logic 1	Logic 1	80 M/S^2	Fig. 4	Fig. 4

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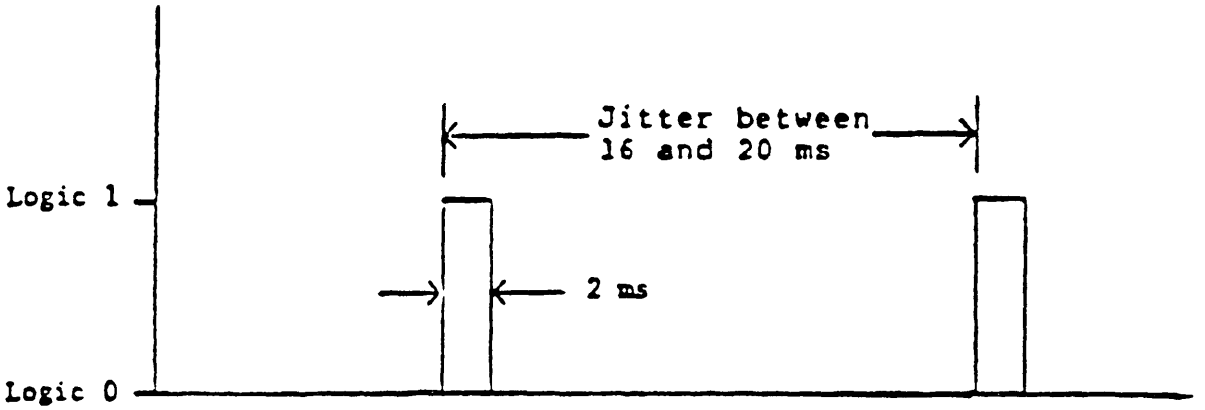


FIGURE 1. Output signal.

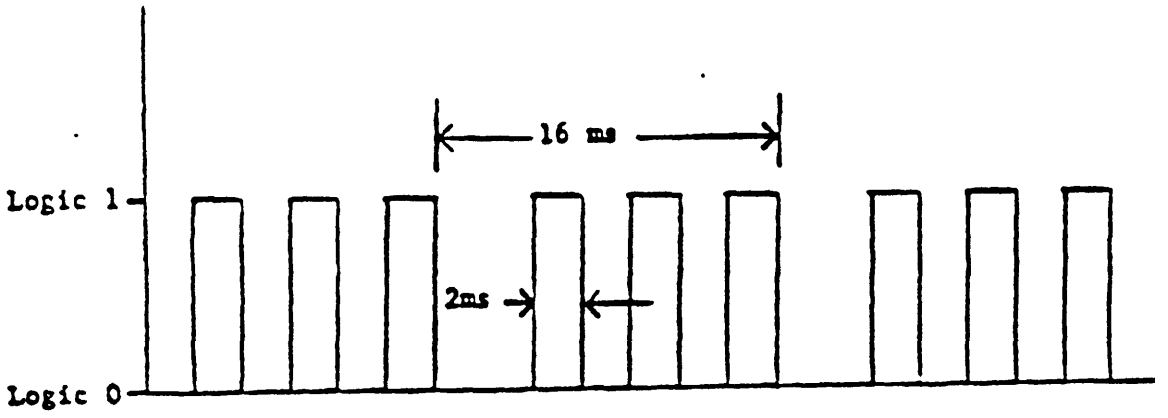


FIGURE 2. Output signal.

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NOTE: τ_1 becomes progressively larger

τ_2 becomes progressively smaller

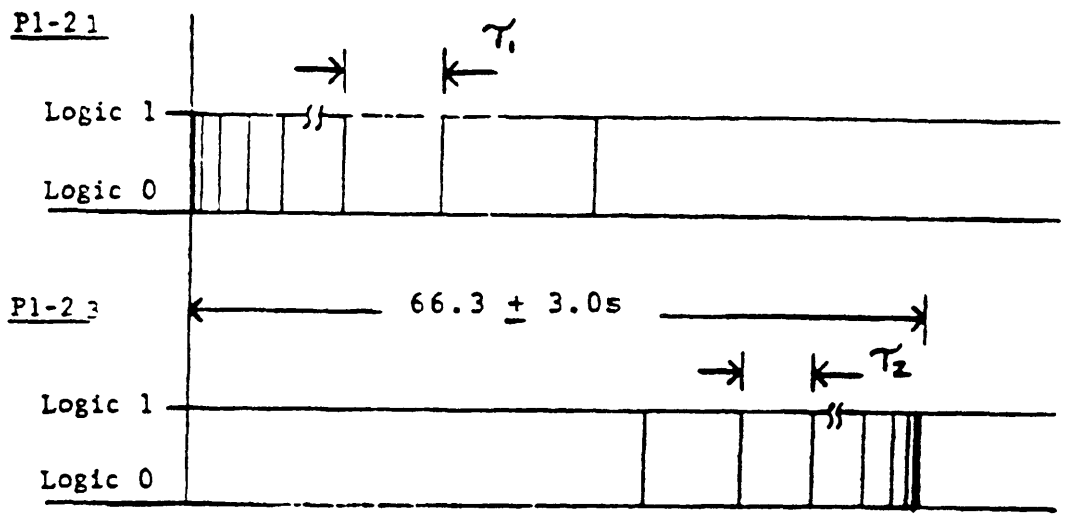


FIGURE 3. Output signal.

NOTE: τ_1 becomes progressively larger

τ_2 becomes progressively smaller

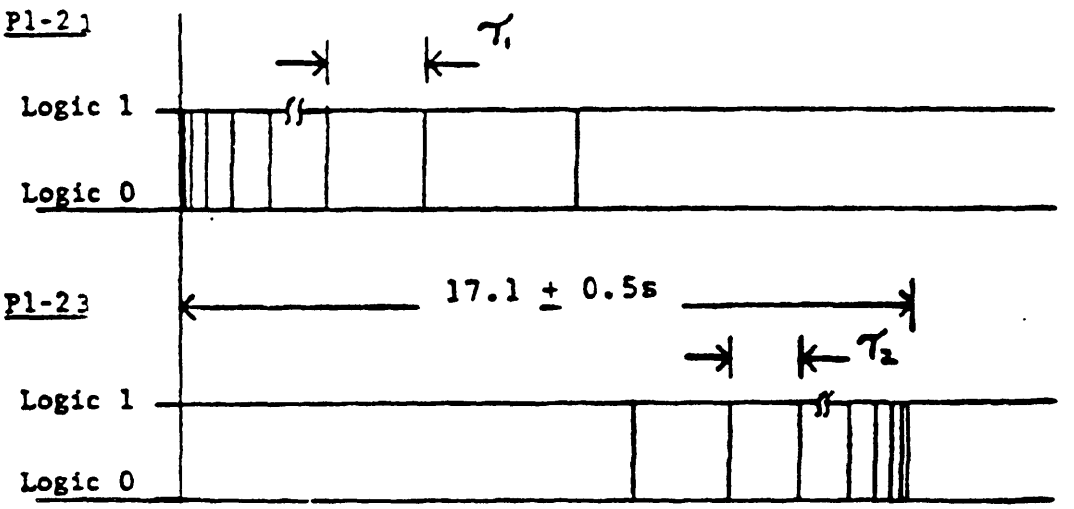


FIGURE 4. Output signal.

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TABLE III. stop , $\overline{\text{stop}}$ characteristics.

	INPUT			OUTPUT		
Test	Pi-l	P1-41	P1-43	P1-52	1i-45	P1-49
1	↓	Logic 1	Logic 1	Logic 1	Logic 1	Logic 0
2	Logic 1	↓	Logic 1	Logic 1	Logic 0	Logic 1
3	↓	Logic 1	Logic 1	Logic 1	Logic 1	Logic 0
4	Logic 1	Logic 1	↓	Logic 1	Logic 0	Logic 1
5	↓	Logic 1	Logic 1	Logic 1	Logic 1	Logic 0
6	Logic 1	Logic 1	Logic 1	↓	Logic 0	Logic 1
7	↓	Logic 1	Logic 1	Logic 0	Logic 0	Logic 1

3.6 First article. When specified in the contract or purchase order see 6.2, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The requirements of MIL-F-13926 shall apply except for soldering.

3.8 Soldering. All soldering shall be in accordance with the general requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance-of all inspection requirements, (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with al-l

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requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated Or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Target Acceleration Boards for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table IV (Functional Test) and Table V (Special Sampling) herein and to any or all requirements of the of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection

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lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with-the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection Plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative-inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government (see 6.3). When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

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4.4.2.3 Functional tests. The tests in Table IV shall be performed on a 100% basis.

TABLE IV. Functional tests.
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical		
102. Performance		
Minor:	None	N/A

4.5 Special sampling. One target acceleration board out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table IV and shall then meet the tests in Table V. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE V. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103 Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production, since the last test as well as to current production, has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table VI shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE VI. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power Supply	5.00 ± 0.25 Vdc, regulated, ac ripple less than 50 mv p-p, 600 ma capacity
2.	Pulse Generator	Positive pulse, O to 4.0 ± 0.5 Vdc, width of 0.2 to 0.6 us at 50% points, 20.000 ± 0.200 KHz repetition rate
3.	Oscilloscope	Dual channel, rise time of oscilloscope plus probe not greater than 10 ns, sweep period less than 50 ns, input resistance 8 megohms or greater
4.	Digital Voltmeter	O to 6 Vdc, .1% accuracy, input impedance greater than 10 megohms
5.	Ohmmeter	o to 1200 ohms, ± 1% accuracy
6.	Test Fixture	As specified in Figure 5. See Table VII for test fixture parts list.

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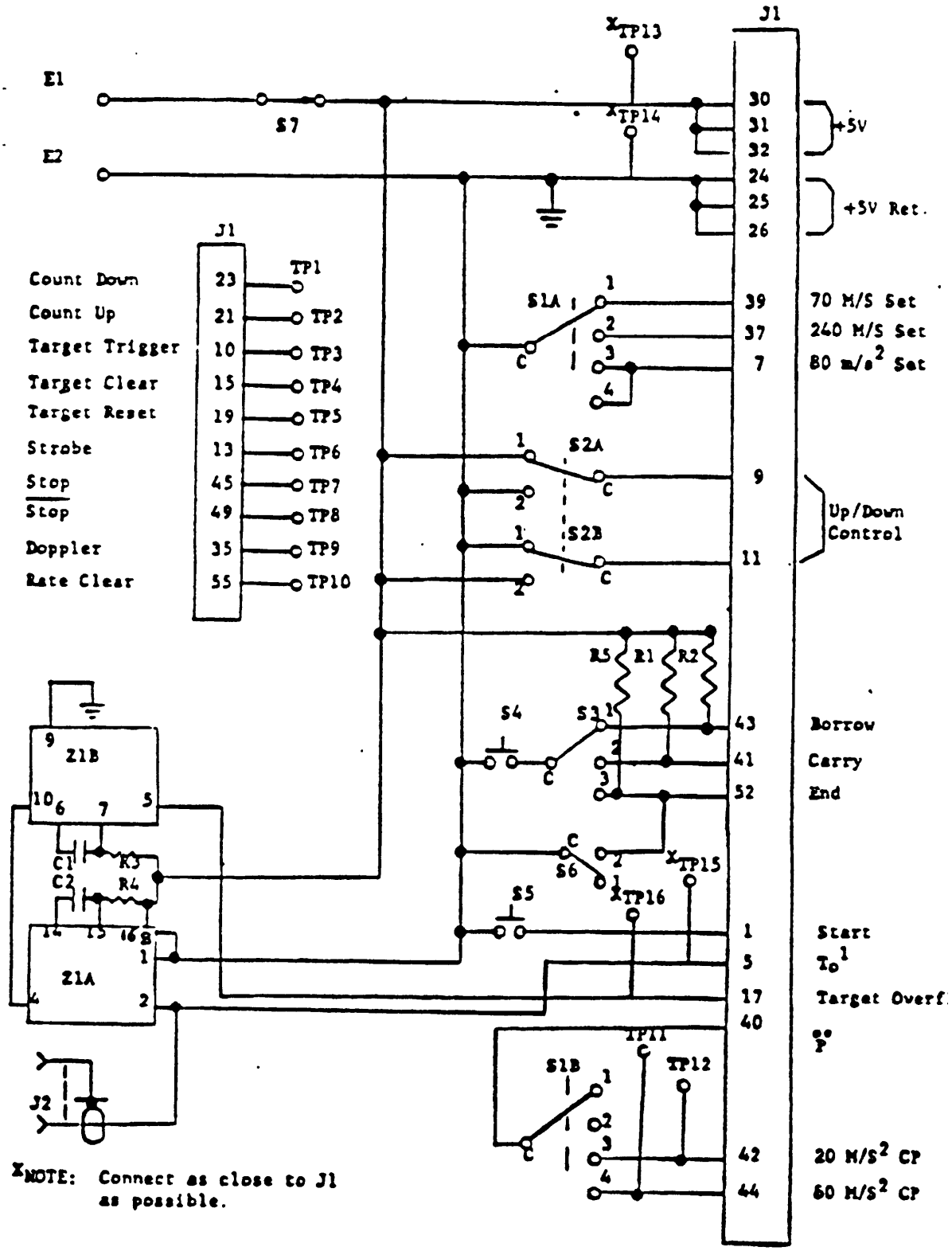


FIGURE 5. Test fixture setup.

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TABLE VII. Test fixture parts list.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
El	Binding post, H. H. Smith 257-R or equiv.
E2	Binding post, H. H. Smith 257-B or equiv.
J1	Connector, per Dwg. 10554938
J2	Connector, BNC receptacle
R1, R2, R5	Resistor, 1K ohms ± 10%, 1/4 watt
S1	Switch two pole four position
S2	Switch DPDT
S3	Switch SPTT
S4, S5	Switch, momentary SPST
S6, S7	Switch, SPST
TP1 thru TP16	Tip Jack, H. H. Smith 1501 or equiv.
Z1	Integrated Circuit Texas Instrument P/N SN54123J or eguiv.
C1,	Capacitor, 470pf ±10%, 100V
C2	Capacitor, 4700pf ±10%, 100V
R3	Resistor, 6.8K ohms ±10%, 1/4 watt
R4	Resistor, 11K ohms ±1%, 1/10 watt

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.701.1 Vibration. This test shall be performed as specified in MIL-T-28800 for Type I, Class 3, Style F test equipment with the exception that Step 4.c Resonance Dwell does not apply. The target acceleration board shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the target acceleration board shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure 1) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms + 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the target acceleration board shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude Operating. This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.,3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 LOW (operating and non-operating) . This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.5 Humidity (non-operatingg This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical.

4.7.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination to determine compliance with 3.4.1.

4.7.2.2 Resistance. Resistance shall be measured with ommeter (item 5) to determine compliance with 3.4.2.

4.7.3 Performance. The selected target acceleration board shall be tested as specified in 4.7.3.1.1 through 4.7.3.1.3 to determine compliance-with 3.5. Logic levels shall be as follows:

Logic O = 0.0 ± 0.5 Vdc

Logic 1 = 4.25 ± 1.00 vdc

↓ = momentary transition from Logic 1 to Logic O

4.7.3.1 Test Setup. Connect target acceleration board (11747207), test fixture (Table VI, item 6), and test equipment (Table VI) as shown in Figure 6.

4.7.3.1.1 Target Signals.

4.7.3.1.1.1 Input power. Use digital voltmeter (Table VI, item 4) to observe +5.00 ± .25 vdc between TP13 (+) and TP14 (Ret) to determine compliance with Input Power requirement of Table 1.

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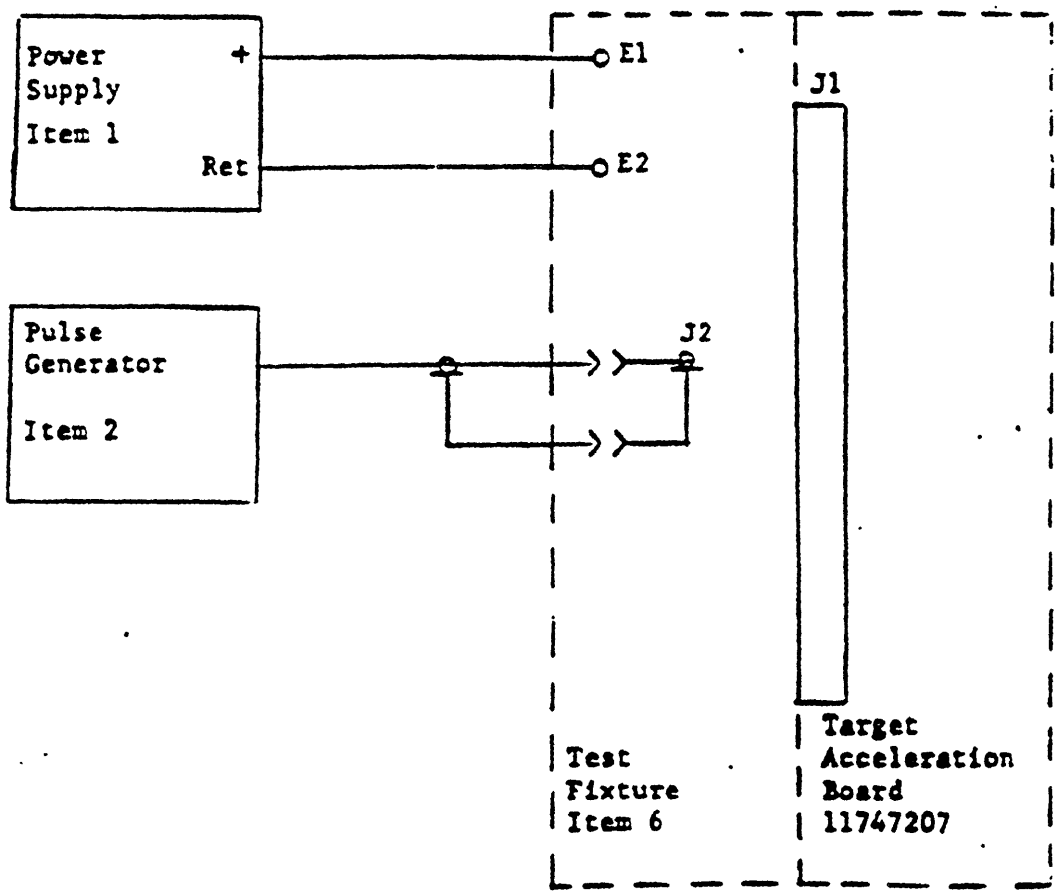


FIGURE 6. Target acceleration test setup.

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4.7.3.1.1.2 Input signals. Use Oscilloscope (item 3) to observe the T_{o}^{1} and Target Overflow signals at TP-15 and TP-16 respectively to determine conformance with Input Signal requirements of Table I.

4.7.3.1.1.3 Output signals. The target output signals specified in Table VIII shall be observed using oscilloscope (Table VI, item 3) at specified test points and with Switch S1 set as specified to determine compliance with output Signal requirements of Table 1.

TABLE VIII. Target output signals

SIGNAL	S1 SWITCH POSITION	TEST POINT
Target Clear	--	TP4
Target Trigger	--	TP3
Target Reset	--	TP5
Strobe	--	TP6
Doppler	--	TP9
80 M/S^2	4	TP11
20 M/S^2	3	TP12

4.7.3.1.2 Count-up, count-down outputs. When the input signals are applied as specified in the below test procedure and Table IX, the output signals in the test procedure and Table IX shall be observed simultaneously using oscilloscope (Table VI, item 3) to determine compliance with 3.5.2. AI-SO, using digital voltmeter (Table VI, item 4) determine that TP10 shall be Logic O during each test, and Logic 1 at any other time.

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TEST PROCEDURE

1. Start test 1 by applying 4 to Pi-l (push S5).
2. Switch S1 to specified position.
3. Switch S2 to specified position.
4. Observe Table IX output signals.
5. Stop test 1 by applying ↓ to P1-52 (set S3 to position 3 and push s4).
6. Observe output signals which shall be at Logic 1.
7. Repeat steps 1 through 6 for tests 2, 3, 4, 5, 6, 7, and 8 respectively.

TABLE IX. Count-up, count-down signals,

	INPUT SIGNALS		OUTPUT SIGNALS	
TEST NO.	S1 POSITION	S2 POSITION	TP2	TP1
1	1	2	Figure 1	Logic 1
2	1	1	Logic 1	Figure 1
3	2	2	Figure 2	Logic 1
	2	1	Logic 1	Figure 2
5	3	2	Figure 3	Figure 3
6	3	1	Figure 3	Figure 3
7	4	2	Figure 4	Figure 4
8	4	1	Figure 4	Figure 4

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4.7.3.1.3 Stop, $\overline{\text{stop}}$ outputs. When the input signals are applied sequentially (Table X-left to right per test, tests 1 through 7) by pushing (↓) or setting switches as specified in Table X, the output signals in Table X shall be observed simultaneously using oscilloscope (Table VI, item 3) to determine compliance with 3.5.3.

TABLE X. stop, $\overline{\text{stop}}$ signals.

	INPUT SIGNALS				OUTPUT SIGNALS	
TEST NO.	S5	S3	S4	S6	TP7	TP8
1	+	--	--	1	Logic 1	Logic 0
2	--	2	↓	1	Logic 0	Logig 1
3	↓	--	--	1	Logic 1	Logic 0
4	--	1	↓	1	Logic 0	Logic 1
5	↓	--	--	1	Logic 1	Logic 0
6	--	3	↓	1	Logic 0	Logic 1
7	↓	--	--	2	Logic 0	Logic 1

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.8.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS1747207.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The target acceleration board is used to provide a simulated target acceleration signal to enable the Range Computer Tester, Subassembly MX-8049/TPM-22, to locate dynamically defective portions of Unit 4 of. Radar Set AN/VPS-2.

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6.2 Acquisition requirements. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a) Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b) Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal-or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC .

6.5 Subject term (Key word) listing.

Fire Control System
Radar Set
Range Computer Tester
Shock Testing
Vibration Testing

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A273)

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